

# # PAPER364; ALICE Multiplicity Centrality: $\rho$ Ratio at Channel 18 and $k_{\eta}$ ; Derivation

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# PAPER364 — ALICE Multiplicity Centrality:  $\rho$  Ratio at Channel 18 and  $k_{\eta}$  Derivation

**Whitepaper Series:** Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF treatment of LHC heavy-ion multiplicity centrality with  $\rho_{SCM}/\rho_{UA}$  ratio at  $n=18$  **Author:** Daniel T. Murphy

<!-- UQFF constants:  $\kappa = 5.0e-4 \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ,  $M_{UQFF} = 1.43e1 \text{ TeV}$  -->

## Abstract

ALICE experiment data on Pb-Pb charged particle multiplicity ( $dN_{ch}/d\eta$ ) vs. collision energy ( $\sqrt{s}$ ) and centrality class are connected to UQFF via the vacuum density ratio at the 18th harmonic channel:  $\rho_{ratio18} = (\rho_{SCM}/\rho_{UA}) \cdot \exp(-[SSq] \cdot 18/26)$ . The empirical multiplicity scaling  $dN_{ch}/d\eta \propto \sqrt{s}^{0.156}$  is reproduced by the UQFF formula with derived coupling constant  $k_{\eta18}$ , connecting heavy-ion collision multiplicities to vacuum-field harmonic structure.

## 2. Core Physics

### 2.1 UQFF Multiplicity Formula

$$\frac{dN_{ch}}{d\eta} = k_{\eta,18} \cdot \left( \frac{\sqrt{s}}{\text{GeV}} \right)^{0.156} \cdot \rho_{ratio,18}$$

where:

$$\rho_{ratio,18} = \frac{\rho_{SCM}}{\rho_{UA}} \cdot \exp\left(-\frac{[SSq]}{26} \cdot 18\right)$$

### 2.2 Vacuum Density Ratio at $n=18$

$$[SSq](18) = e^{-0.57 \cdot 18/26} = e^{-0.394} \approx 0.674$$

$$\rho_{ratio,18} = 0.1 \cdot 0.674 = 0.0674$$

### 2.3 Empirical Scaling Exponent

The ALICE measured scaling for Pb-Pb at 0–5% centrality:

$$\frac{dN_{ch}}{d\eta} \Big|_{\text{central}} \approx 5.28 \cdot \left( \frac{\sqrt{s}}{1 \text{ GeV}} \right)^{0.156} \cdot \frac{N_{part}^2}{2}$$

UQFF prediction:

$$k_{\eta,18} \cdot \rho_{ratio,18} = 5.28 \cdot \frac{N_{part}^2}{2} \implies k_{\eta,18} = \frac{5.28}{0.0674} \cdot \frac{N_{part}^2}{2}$$

### 2.4 $k_{\eta18}$ (Derived Coupling Constant)

For central Pb-Pb ( $N_{\text{part}} \sim 380$ ):

$$k_{\{\eta,18\}} \approx \frac{5.28}{0.0674} \times 190 \approx 14,887$$

This coupling constant quantifies the UQFF vacuum field's contribution to the final hadron multiplicity in ultra-relativistic heavy-ion collisions.

### 3. Key Values

| Quantity                    | Formula                            | Value            |
|-----------------------------|------------------------------------|------------------|
| Scaling exponent            | $dN/d\eta \propto \sqrt{s}^\alpha$ | $\alpha = 0.156$ |
| $[SSq](18)$                 | $\exp(-0.57 \cdot 18/26)$          | 0.674            |
| $pratio18$                  | $0.1 \times [SSq](18)$             | 0.0674           |
| $k\eta18$                   | Derived from ALICE                 | $\sim 14,887$    |
| $N_{\text{part}}$ (central) | ALICE Pb-Pb                        | $\sim 380$       |

### 4. Physical Significance

This paper bridges UQFF to the most precision heavy-ion physics dataset available. The choice of  $n = 18$  channel is physically motivated: in a system containing 26 quark/gluon vacuum layers ( $\Sigma \blacksquare$ ), the 18th channel corresponds to the transition between perturbative ( $n < 13$ ) and non-perturbative ( $n > 13$ ) QCD regimes. The UQFF prediction that multiplicity scales as  $pratio18$  is testable across the full collision energy range ( $\sqrt{s} = 2.76$  to  $5.36$  TeV) — the  $k\eta18$  value should remain constant if UQFF is correct.

### 5. Deduplication Note

**\*\*vs. PAPER\_364 vs. earlier LENR/nuclear papers:\*\*** PAPER\_340 and PAPER\_351 treat nuclear processes in astrophysical contexts; PAPER\_364 is the first UQFF calculation for collider heavy-ion experiments.

**Unique:** LHC ALICE data;  $dN_{\text{ch}}/d\eta$  centrality scaling is unique to this paper.

### 6. Classification

**Physics Territory:** FIRST UQFF LHC heavy-ion multiplicity centrality with  $p_{\text{vac}}$  ratio at  $n=18$  channel

**Scale:** Sub-nuclear (heavy-ion collision,  $\sqrt{s}$  = several TeV) **CP Implementation:**

ALICEMultiplicityCentralityRhoVacRatioCalculator (CondensedPhysics4.py, Session 97)